

"RCC Work estimation in Superstructure"

Workout the quantities of steel or RCC beam used over a clear span of 5.5M. The walls supporting the beam has 300mm bearing over the wall on both sides. The size of beam 250x550mm concrete cover at the ends of bars and sides in 40mm and top and bottom 30mm each.

i) Main reinforcement @ bottom -22 mm diameter, 2 Numbers

ii) Main bent up bars- 22mm diameter, 2 Numbers.

iii) Top bars- 16MM diameter, 2 Numbers.

iv) Stirrups-8mm diameter @ both 1.5m, 150mm c/c-2 Numbers. 16mm diameter @ middle 2.5M, 210mm c/c-2 numbers

Bearing =	300	mm
length @ both ends =	1.5	M
Length @ middle =	2.5	M
Number of bar at all sides =	2	Numbers
Length of the beam =	250	mm
Breadth of the beam =	550	mm
Cover @ top and bottom =	30	mm
Cover @ sides =	40	mm
Diameter of the Main reinforcement =	22	mm
Main bent up bars =	22	mm
Top bars of the diameter =	16	mm
Stirrups diameter =	8	mm
Distance between stirrups centre to centre @ stirrups =	150	mm c/c
Distance between stirrups centre to centre @ middle =	210	mm c/c

i) Main Reinforcement of Bar.		
Length of the bar =	clearspan + Bearing - end cover	
Length of the bar =	6.02	M
Totall length of the bar =	length of bar + 2 hooks	
Hooks =	2x9x diameter of the bar	
Totall length of the bar =	6.416	M

ii) Main bent up bars		
Length of the bar =	6.02	M
Total length of the bar =	length of the bar + 2x hooks+2x0.42xdepth.	
depth =	depth of the beam - top and bottom cover+ diameter of bar.	
depth =	468	mm
Total length of the bar =	6.80912	M

iv) Stirrups		
Length of stirrups =	2*[(B+D)]-2xcover top and bottom-2xSide cover+2x12 diameter of bar	
Length of stirrups =	1.512	M
Number of stirrups @ one end =	11	Number
Number of stirrups @ other end =	11	Number
Number of stirrups @ middle =	10.9048	Number
Number of stirrups @ middle =	11	Number
Total number of stirrups =	33	Number

iii) Distribution bar.		
Length of the bar =	6.02	M
Main bar total length =	6.416	M
Distribution total length of the bar =	Length of the bar + 2x9xdiameter.	
Distribution total length of the bar =	6.308	M

[illegible]